Motor control exercise

1. After amputation of a limb, many patients report that they still feel the presence of the limb, including sensations such as movement, pain, and itching – a phenomenon known as a phantom limb. Some patients report being able to move their phantom limbs voluntarily, while others experience the phantom as paralyzed and cannot move it even with intense effort. From the perspective of motor control, this phenomenon cannot be solely attributed to the sensory information received. Describe, in the context of internal models of motor control, what may be the cause of phantom limbs.

2. Imagine you got talked into going skydiving, but when it was time to jump, you realized you had maybe been impulsive and needed to rethink this decision. Given that you are anxious person, who tends to overestimate risk, you have the impression that the probability of something going wrong during the skydive is about 0.1. Being aware of your risk overestimation, you decide to look up the statistics. You take out your phone and google the probability of dying when skydiving. Out of 3.3 million skydives, there were 15 fatalities. However, in making the decision whether to skydive or not, you also have to take into account your losses. Skydiving is exhilarating, and you have wanted to try it your whole life, so say the loss of having a positive experience is denoted as -1 (negative reflects pleasure). Of course, dying would be very bad, so the loss of having an accident is say 100,000. Do you jump?

3. Let's apply decision theory to a sensorimotor control example. Say you have to move your hand as fast as possible to reach a certain target, and you have two chances (two movements) to do so. You either a) "choose" to make the 2 movements where both miss the target by 2 cm, or b) the first misses it by 4 cm, and the second hits the target. Your loss function specifies which choice is better. Which loss function might you have if choice a) is better, and which if choice b) is better? and which if both choices are equally good? (e.g. linear, quadratic, etc.?)

4. You are reaching to a target along a horizontal line. Your **prior belief** about the target's position (in cm, relative to the midline) is Gaussian with mean $\mu_p = 0$ and variance $\sigma_p^2 = 4$. On this trial, your **visual observation** of the target is $z = 3$ cm with visual noise variance $\sigma_\ell^2 = 1$.

1. Compute the posterior mean and variance for the target position.
2. Now suppose the visual feedback becomes much noisier, $\sigma_\ell^2 = 9$. Recompute the posterior mean and variance.
3. Briefly explain how this illustrates Bayesian integration in sensorimotor control.